AA272 - Beyond EKF: evaluation and application of advanced filters

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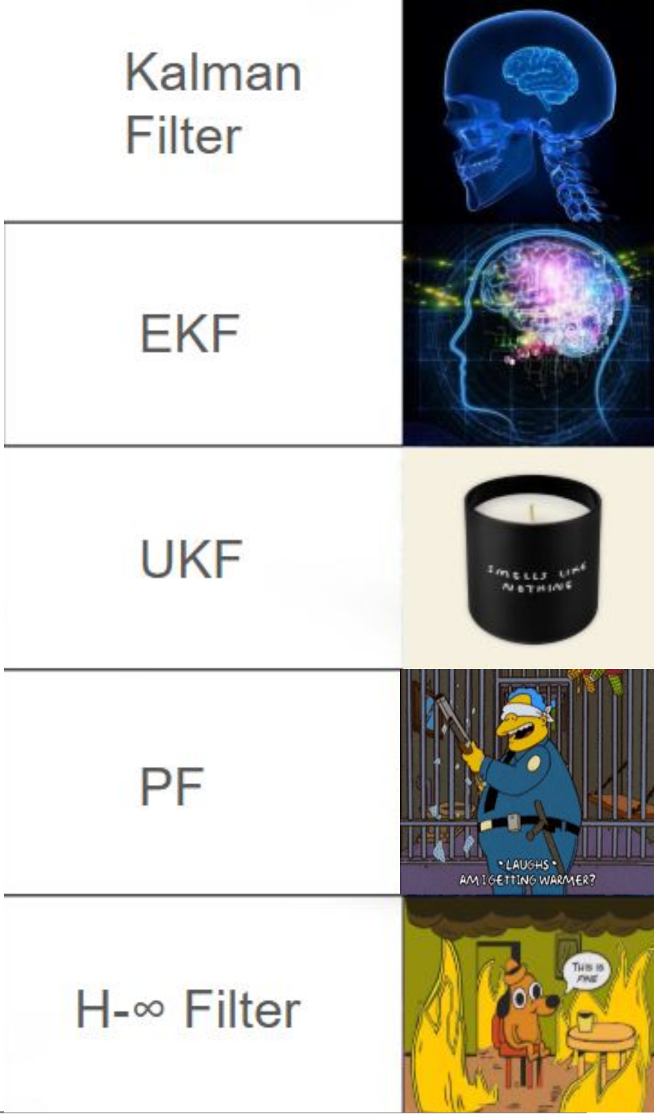
Introduction

- Original Project Idea: State Augmentation

- Modified Project: Advanced Filtering Suite
  - Different filters = different assumptions & applications
  - UKF, PF, H-∞ Filter

- State of the Art: EKF (built-in gnss-lib-py solver)
  - Nonlinear F and H
  - Mean-zeroed Gaussian Noise
  - linearize => predict => update
  - all-around candidate



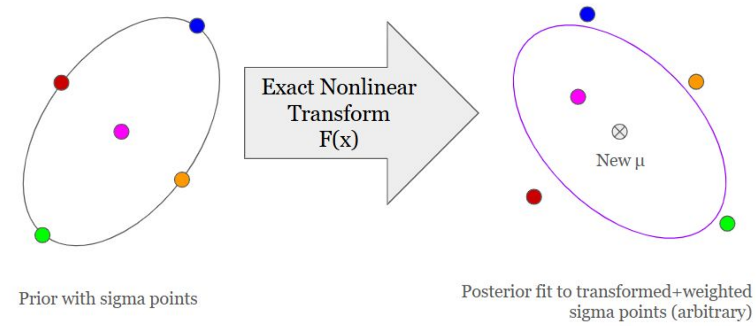
Preview: Filter Comparative Summary

Filter	F(x), H(x)	Noise Assmp.	Runtime	Optimality	Best For?
KF	Linear	μ-zeroed Gauss	Low	Deterministic, **Provably Optimal**	Exact linear systems
EKF	Nonlinear (1 linearized)	μ-zeroed Gauss	Low	Deterministic	**Mildly nonlinear and differentiable systems**
UKF	**Exact Nonlinear**	μ-zeroed Gauss	Moderate	Deterministic, **Hyper-parameter dependent**	Highly nonlinear or discontinuous systems
PF	Exact Nonlinear	**None**	High	**Stochastic**, N and reselection-dependent	multi-modal noise
H-∞	Nonlinear (1 linearized)	**None**	Low (EKF comparable)	Deterministic, Constrained Optimality under hyper parameterized noise limit γ	**Worst-case** robustness, adversarial agents

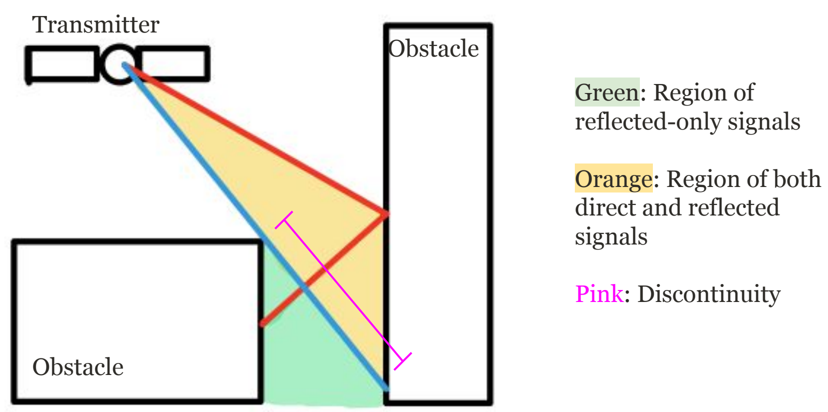
Algorithm 1 - Unscented Kalman Filter (UKF)

- Intuition:
  - **No linear approximations**
  - Propagate **Sigma Points**
    - points to best capture a posterior Gaussian distribution
  - 2x "Unscented" Transforms
- Key Assumptions:
  - Nonlinear dynamics/measurements
  - **Unimodal** Gaussian noise
- Use-Cases:
  - Sensor-educated dynamics
  - Jacobian breakdowns
    - Urban canyon NLOS reflection discontinuities

Example of Unscented Transform



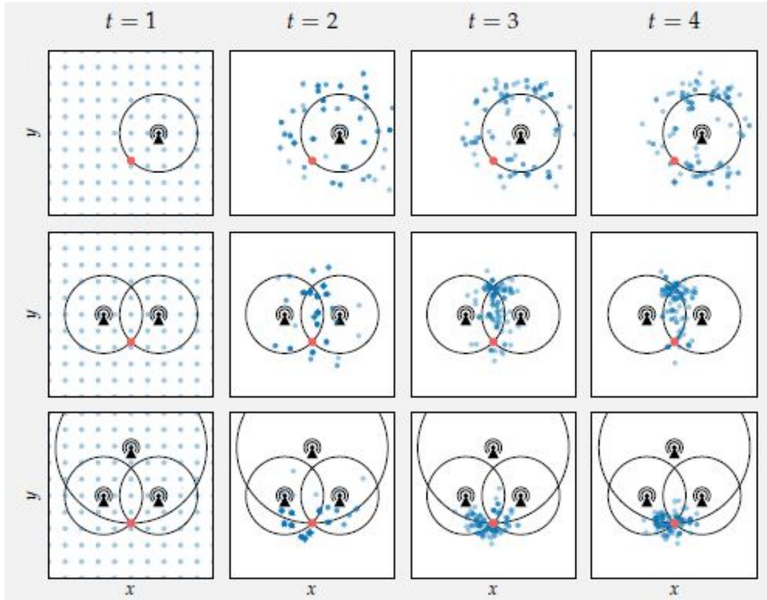
Example 2D of discontinuity (breakdown of Jacobian)



Algorithm 2 - Particle Filter (PF)

- Intuition:
  - **Stochastic**, and scaled-up flavor of UKF
  - Propagate many weighted particles through models as an estimate of posterior
    - convergence by reweighting and elitist selection
- Key Assumptions:
  - Nonlinear dynamics/measurements
  - **No noise assumptions**
- Use-Case:
  - Urban Canyons
  - Carrier Phase offset integer ambiguities

Particle Filter Visualization for 2D beacon positioning



Kochenderfer, Mykel J., et al. *Algorithms for Decision Making*. The MIT Press, 2022, p 391

Optimal under worst-case inequality constraint.

Algorithm 3 - H-∞ Filter

- Intuition:
  - **Robust EKF (γ robustness parameter)**
    - aims to keep "total disturbance energy" below a hyper parameterized threshold
  - Low γ: stiff, robust to disturbance
    - risks infeasibility by non invertibility of S
  - High γ: elastic, weak robustness
    - approaches EKF as γ -> ∞
- Key Assumptions:
  - No noise assumptions (agnostic to Q and R from EKF)
  - Nonlinear dynamics/measurements
- Use-Case (niche):
  - **Adversarial agents (spoofers/jammers)**
  - One-off satellite errors

H-∞ Filter aims to satisfy the following inequality

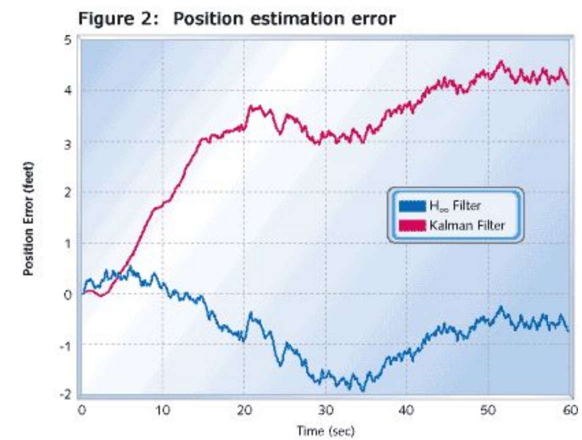
$$\sum_{t=0}^{N} \|x_t - \hat{x}_t\|^2 \leq \gamma^2 \sum_{t=0}^{N} \left(\|w_t\|^2 + \|v_t\|^2\right)$$

Modified EKF innovation and gain step

$$S_t = H_t P_t^- H_t^T - \gamma^{-2} I$$

$$K_t = P_t^- H_t^T S_t^{-1}$$

Visualization of H-∞ Filter robustness vs KF



source: https://www.embedded.com/from-here-to-infinity/

Recap: Filter Comparative Summary

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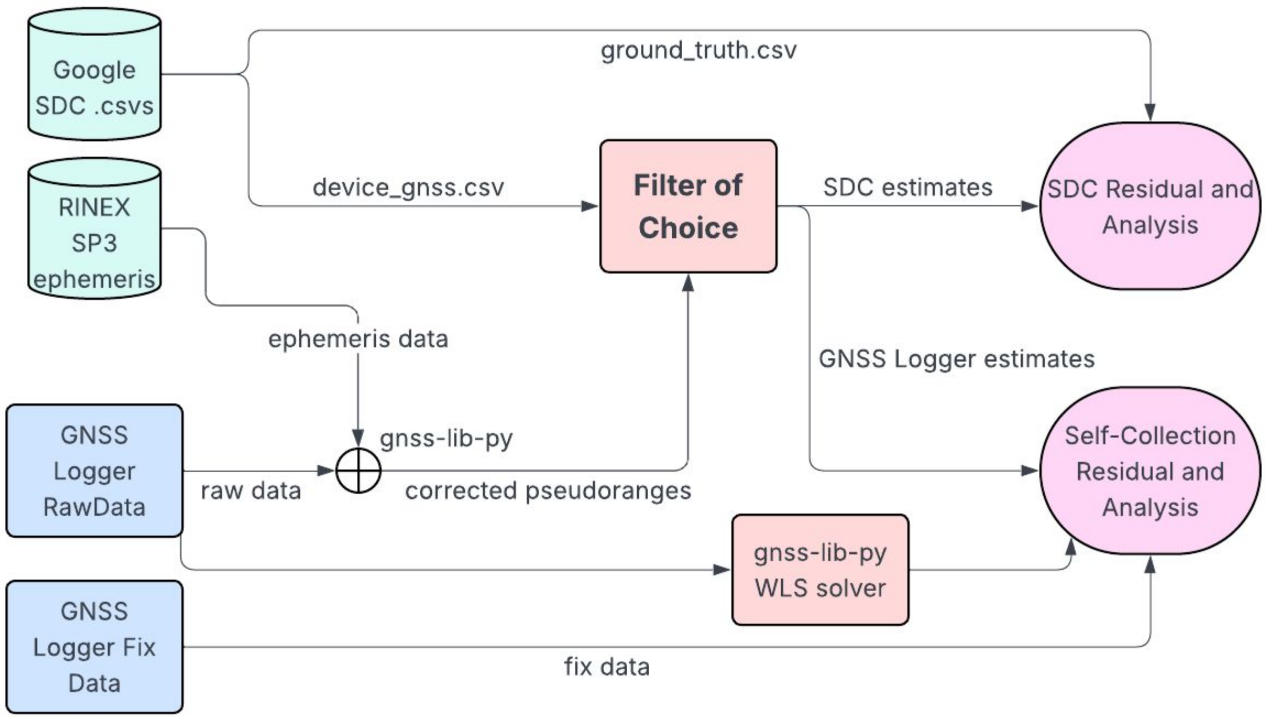
Methods



RMSE Residuals: Estimated to Ground Truth cumulative errors

Runtime

Qualitative Comparisons: Trajectory behavior

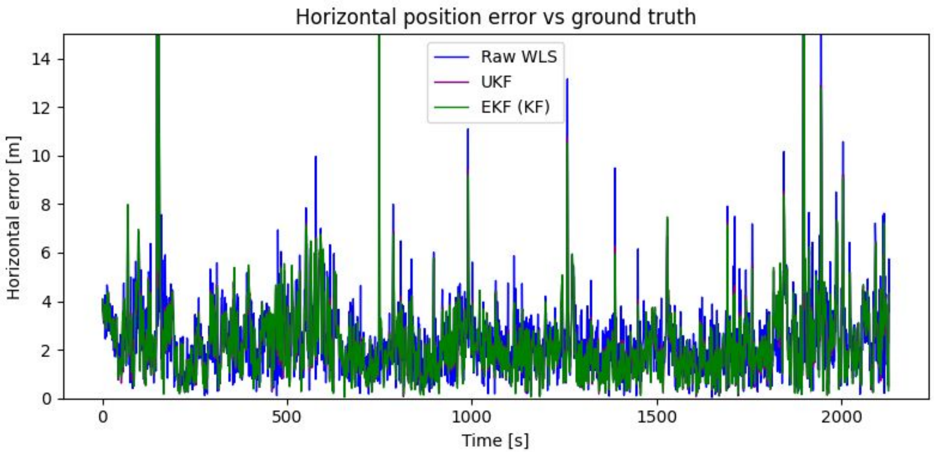
Results Summary

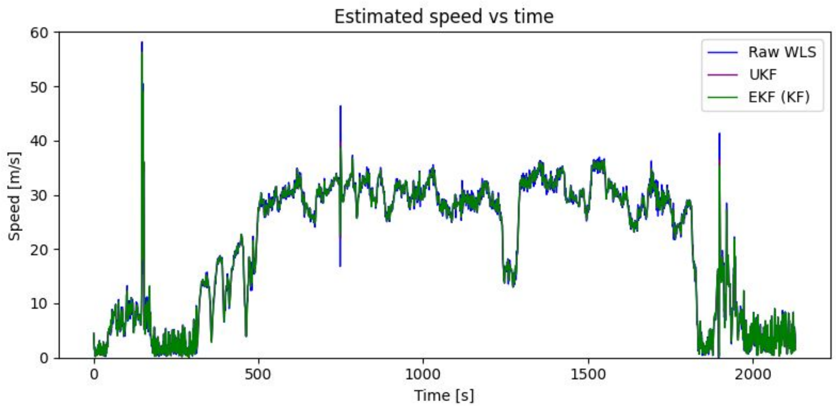
Filter	2D Position RMSE [m]	Relative 2D Position RMSE	Runtime [ms]	Relative Runtime
WLS Solution	3.245	1.155	144.87	1
EKF	2.915	1.036	176.13	1.215
UKF	2.811	1	838.62	5.787
PF (N = 5000)	17.582	6.254	3199	22.08
H-∞*	2.994	1.065	867.65	5.991

Lowest RMSE, Moderate Runtime

$\boldsymbol{\alpha}$	**0.1**
β	2
κ	0

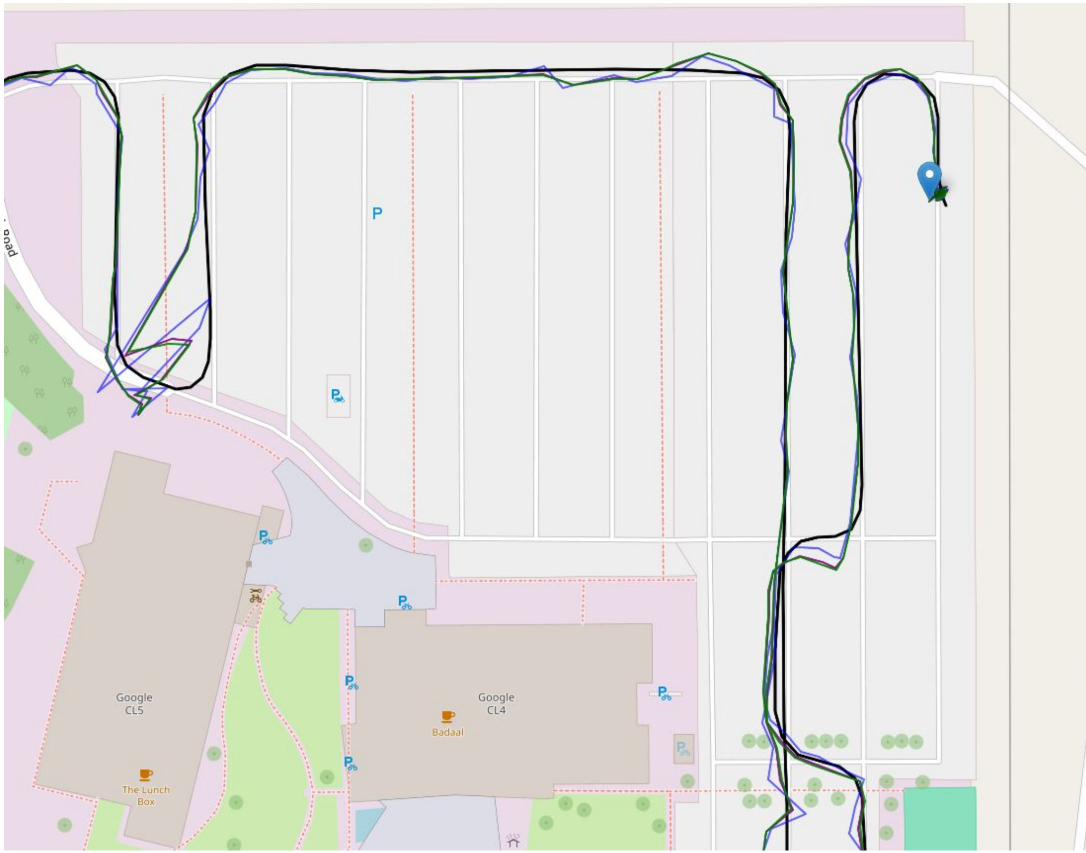
UKF Analysis





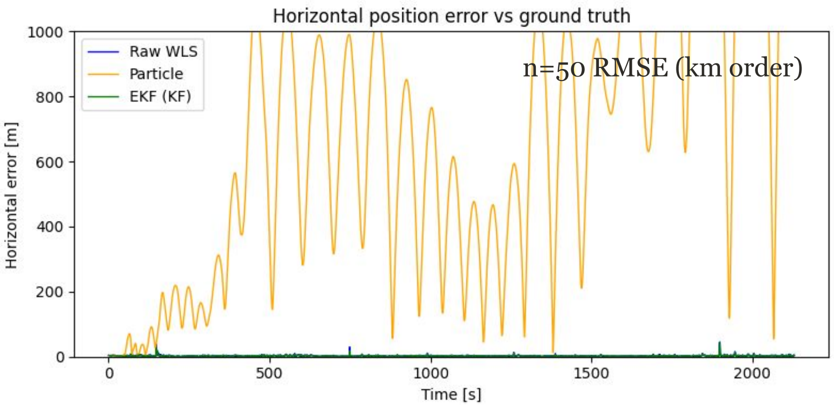
α	**0.1**
β	2
κ	0

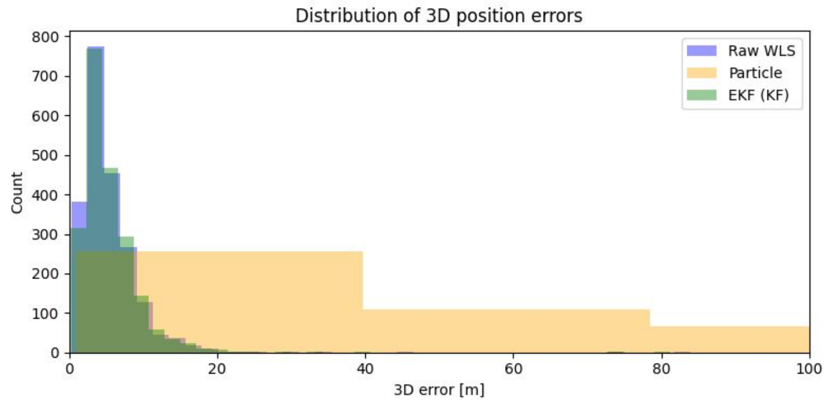
UKF Trajectory

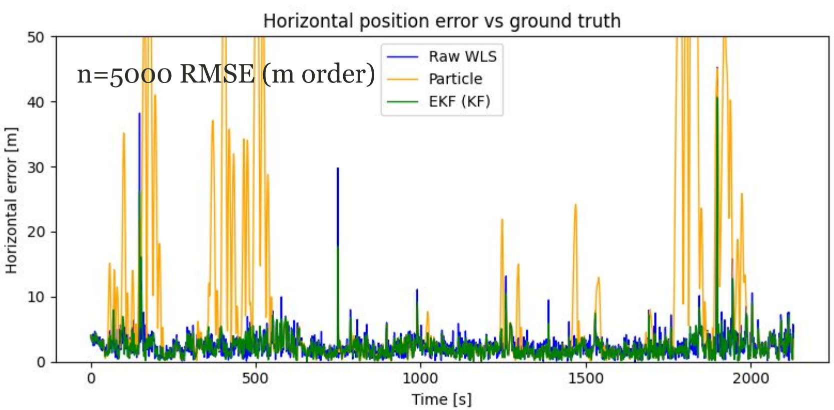


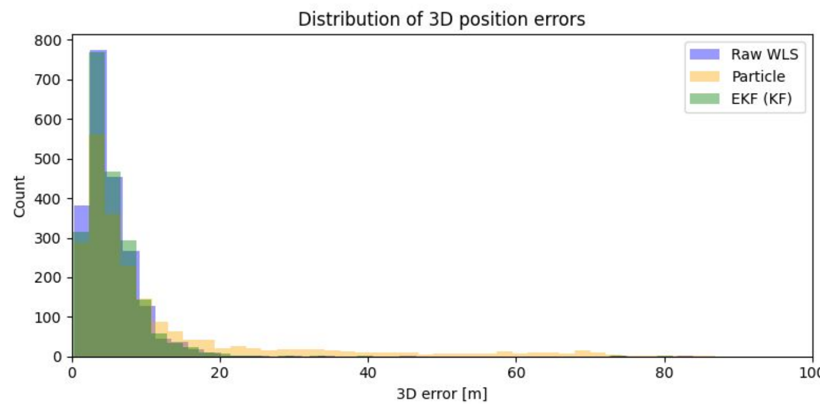
- Similar Trajectory to EKF
- nonlinearities are not extreme.
- Still marginal improvement

PF Analysis

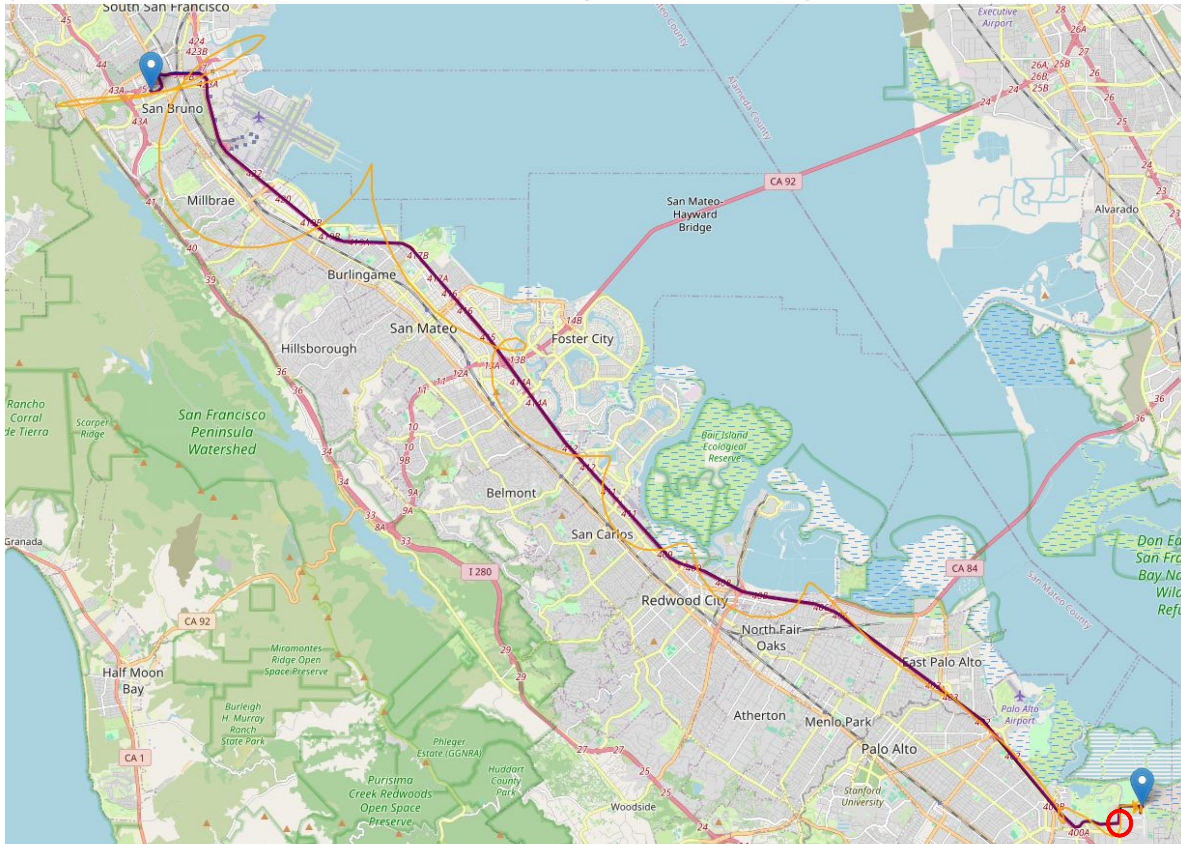

Horizontal position error vs ground truth
Raw WLS
Particle
EKF (KF)
n=50 RMSE (km order)
Horizontal error [m]
Time [s]


Distribution of 3D position errors
Raw WLS
Particle
EKF (KF)
Count
3D error [m]


Horizontal position error vs ground truth
n=5000 RMSE (m order)
Raw WLS
Particle
EKF (KF)
Horizontal error [m]
Time [s]


Distribution of 3D position errors
Raw WLS
Particle
EKF (KF)
Count
3D error [m]

PF Analysis (N = 50 trajectory)

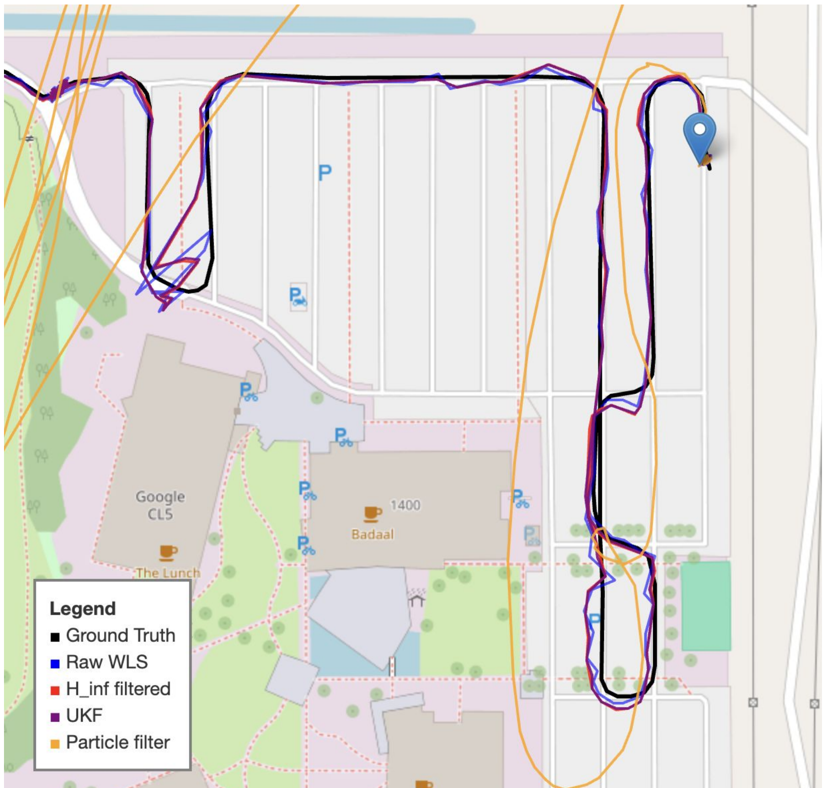


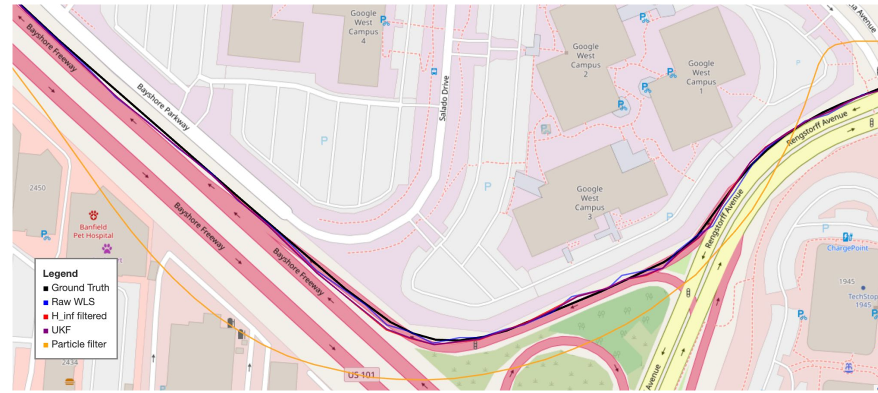
High RMSE, Extreme Runtime
*Runtime and RMSE inversely correlated

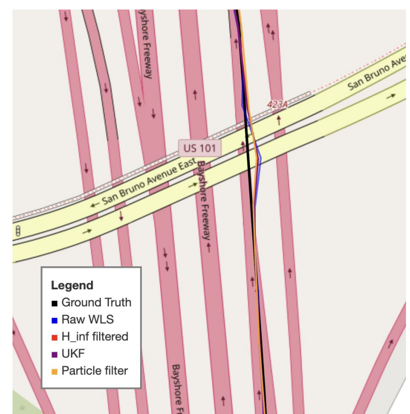
PF Analysis (N = 5000 trajectory)

H-∞ Analysis

𝝈_sigma = 1.0	Measurement disturbance energy: small meas_sigma -> we trust the raw GNSS more
𝝈_accel = 0.5	Model disturbance energy: Small 𝝈_accel -> smoother model, assuming constant acceleration
γ = 5*10^-4	mildly robust, ensures the estimation error energy stays below γ^2

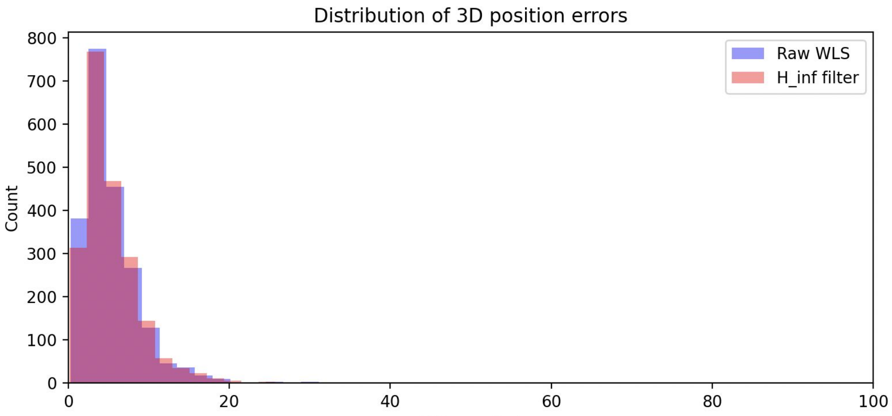




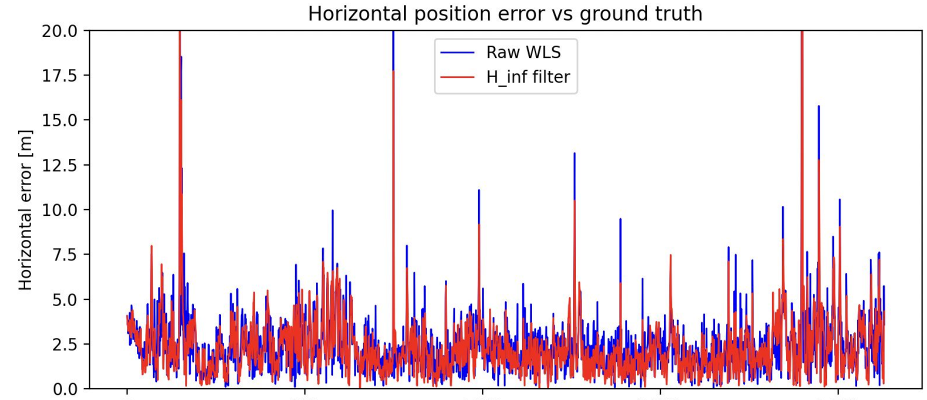


- Similar Trajectory to UKF and WLS
- Great tracking in smooth areas
- Acceptable tracking in coarse areas

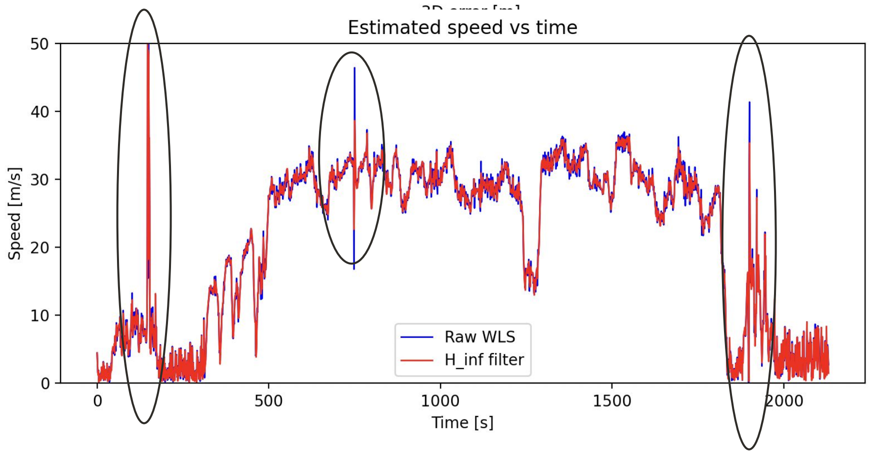
H-∞ Analysis



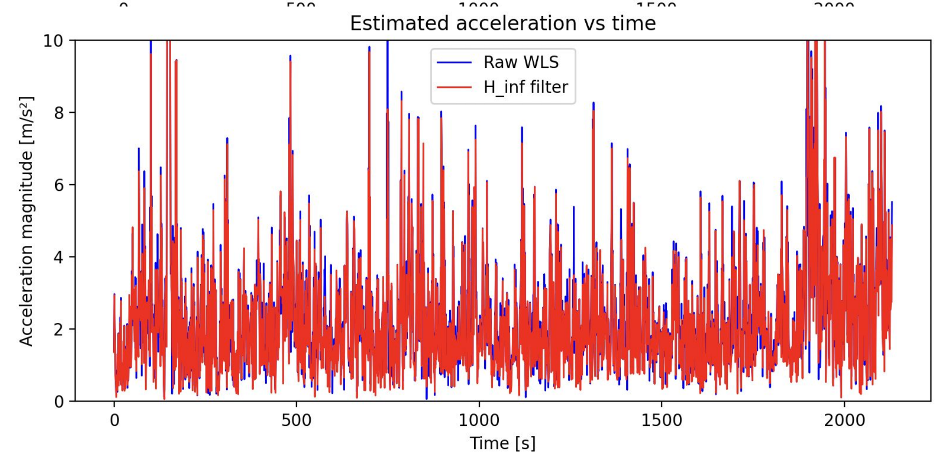
Distribution of 3D position errors
Raw WLS
H_inf filter
Count
0
100
200
300
400
500
600
700
800
0
20
40
60
80
100



Horizontal position error vs ground truth
Raw WLS
H_inf filter
Horizontal error [m]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



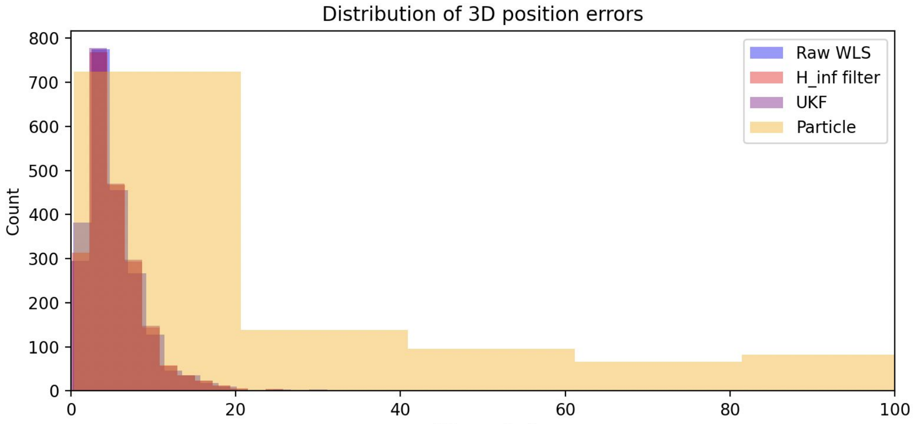
Estimated speed vs time
Raw WLS
H_inf filter
Speed [m/s]
0
10
20
30
40
50
0
500
1000
1500
2000
Time [s]

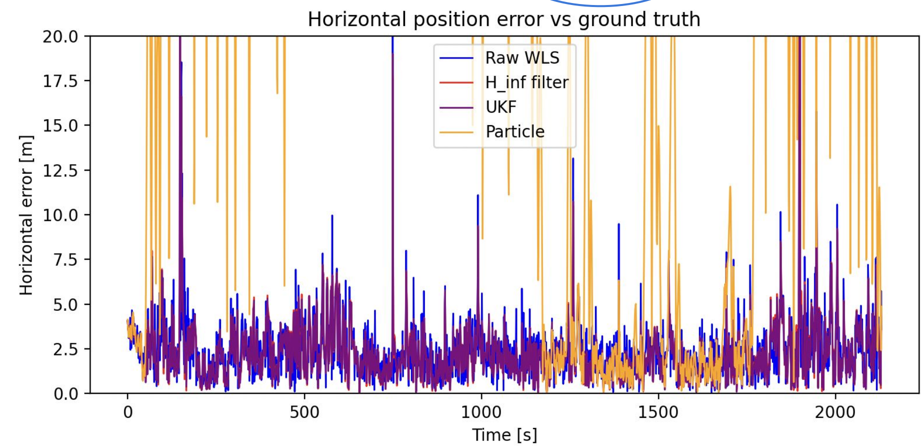


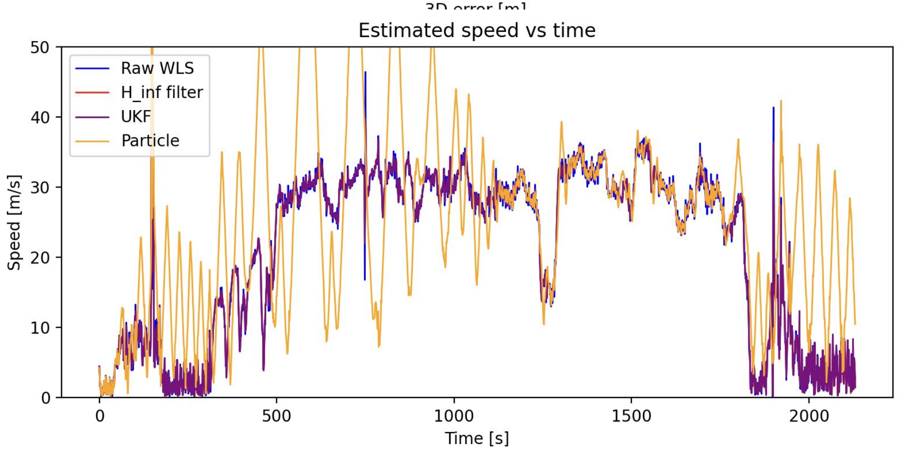
Estimated acceleration vs time
Raw WLS
H_inf filter
Acceleration magnitude [m/s²]
0
2
4
6
8
10
0
500
1000
1500
2000
Time [s]

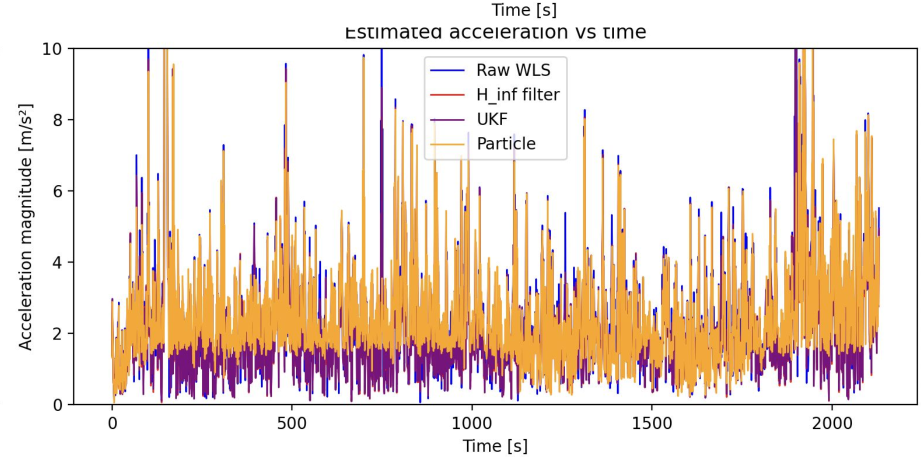
H-∞ Analysis (Bigger Picture)

Similar performance to UKF at the same compute time!



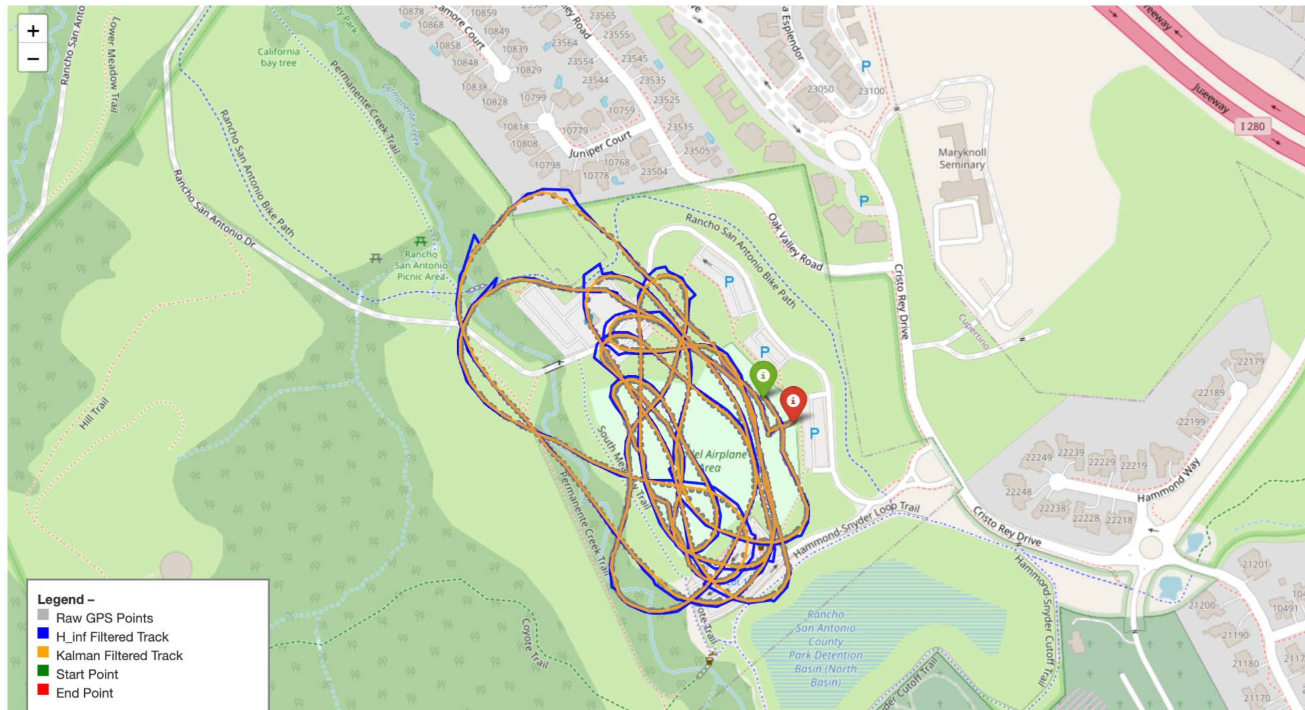




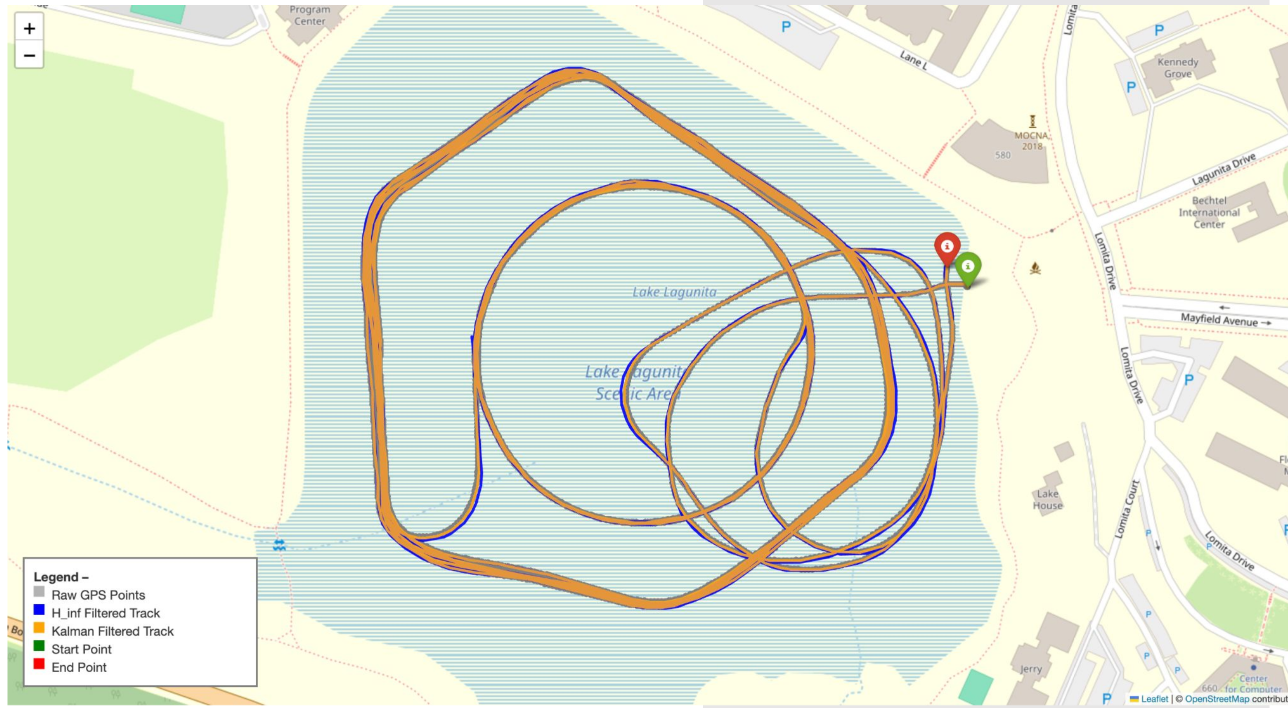


H-∞ Filter, Extreme Case

Coyote Hill Test Flight
No Ground Truth
On-board GPS



H-∞ Filter, Gentle Case



Coyote Hill Test Flight
No Ground Truth
On-board GPS

Future Recommendations

1. New or hybrid filter implementations
    a. Adaptive EKF - dynamic Q and R
2. Sensitivity study on filters with hyperparameters
3. Evaluation of filter performance in environments with pre-characterized noise/biases and spoofers/jammers
    a. Certain filters require niche noise and adversarial environments to show their true effectiveness over the EKF
    b. PF and H-inf Filters respectively
    c. Artificial? - begs the question of how to replicate truth

Acknowledgements

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